

# Fixed-Time Fuzzy Control of Uncertain Robots With Guaranteed Transient Performance

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**Abstract**—In this article, an adaptive fixed-time fuzzy control scheme is proposed for an uncertain robot manipulator with user-defined performance. A novel symmetrical barrier Lyapunov function is designed based on the error conversion mechanism and the performance function such that the tracking errors will not violate the prescribed output constraints. A novel adaptive law is constructed and incorporated into the fixed-time controller design such that all the closed-loop signals can be bounded and achieve practical fixed-time convergence regardless of the initial conditions. Finally, the feasibility and superiority of the proposed scheme are demonstrated based on simulation and experimental studies using a Baxter robot.

**Index Terms**—Adaptive fuzzy control, barrier Lyapunov function (BLF), fixed-time convergence, robot manipulator.

## I. INTRODUCTION

FOR the robot dynamic systems, uncertain nonlinear terms usually exist due to the time-varying model parameters and the external disturbance during operation. Combined with adaptive control techniques, the neural network (NN) [1]–[5] and the fuzzy logic system (FLS) [6]–[8] have been widely applied to handle the tracking control problem for the uncertain robot systems because of their universal approximation capability. In [4], an NN control scheme has been proposed for a robot manipulator to achieve trajectory tracking with output constraints and input saturation. In [5], an admittance adaptation method has been proposed for robot–environment interaction, and the

guaranteed trajectory tracking can be achieved by an NN-based controller. In [9], a number of advanced NN control algorithms have been introduced in detail for nonlinear systems, including robots. In [7], an adaptive robust fuzzy control scheme has been proposed for the uncertain two-degree-of-freedom (DOF) lower limb exoskeleton robot system to enhance the rehabilitation training. In [8], an adaptive fuzzy control scheme has been proposed for robotic systems to achieve fixed-time convergence and user-defined performance simultaneously. In [10], a novel adaptive controller based on active inference has been proposed for the robot to handle large model uncertainties and a large number DOFs, which enable to deal with the robot uncertainty, besides the NN and the FLS. It should be noted that large calculation is always required because of the weight iterative process in traditional fuzzy control schemes. In our previous work [11], a topology optimization method has been proposed to reduce the data amount for calculation and storage in wireless sensor networks, which is exciting because it may enlighten optimizing the FLS structure to reduce the computational complexity in our future work. Moreover, the desired transient performance and convergence time are rarely discussed simultaneously in most of the existing fuzzy control schemes.

In practice, the undesirable transient performance may lead to the system instability, even the system safety problems sometimes. Recently, the barrier Lyapunov functions (BLFs) have been widely used to achieve the state and output constraints in the nonlinear control problems [12]–[16]. In [12], with the exponential-type BLF, a practical event-triggered prescribed-time controller has been proposed for a class of space teleoperation systems. In [14], a new command filtered fuzzy controller has been proposed for a class of unknown nonlinear systems to handle full-state constraints and finite-time convergence simultaneously. In [16], an adaptive fuzzy leader-following tracking control scheme has been proposed for heterogeneous nonlinear multiagent systems with finite-time output constraints. In this article, a novel symmetric BLF is designed to guarantee the desired transient performance of the robot system.

In many industrial systems, the system states are required to achieve fast convergence speed for better control performance. There have been some proposed research works focused on the convergence time of the systems [17]–[19]. In [17], an adaptive observer-based fuzzy controller has been proposed for a class of strict-feedback nonlinear systems to achieve finite-time convergence. In [18], an adaptive finite-time sliding-mode control scheme has been proposed for a class of nonlinear systems

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with some matched uncertainties. Nevertheless, for the existing finite-time control schemes, the convergence time of the systems is always related to the initial conditions, which are sometimes unavailable. To improve the control performance, the fixed-time control schemes have been proposed and applied in the nonlinear control community [20]–[22]. In [20], a novel fixed-time adaptive fuzzy control scheme combined with the BLF technique has been proposed for uncertain nonstrict-feedback nonlinear systems. In [21], an adaptive event-based fixed-time control scheme has been proposed for the active vehicle suspension systems, and the predefined constraints can be guaranteed.

Motivated by the above research works, the problem of fixed-time tracking control is discussed for uncertain robot systems based on the FLS and the BLF technique in this article. The major contributions of our work can be listed as follows.

- 1) A novel symmetric BLF is designed to avoid the violation of the output constraints; thus, the desired transient performance of the robot system can be guaranteed.
- 2) A novel adaptive law is proposed such that the boundedness of all the closed-loop signals can be proved. Then, the assumption that the weight estimation is bounded in recent fixed-time control research [23]–[25] can be relaxed.
- 3) The tracking performance of the robot system can achieve practical fixed-time convergence regardless of the initial conditions.

## II. PROBLEM FORMULATION AND PRELIMINARIES

### A. Problem Formulation

The following dynamic equation of an  $n$ -DOF robot is considered:

$$M(q)\ddot{q} + C(q, \dot{q})\dot{q} + G(q) + \tau_d(q, \dot{q}) = \tau \quad (1)$$

where  $q, \dot{q}, \ddot{q} \in R^n$  are expressed as the vectors of position, velocity, and acceleration, respectively.  $M(q), C(q, \dot{q}) \in R^{n \times n}$ , and  $G(q) \in R^n$  mean the unknown inertia matrix, centripetal Coriolis torque matrix, and gravitational vector, respectively.  $\tau \in R^n$  and  $\tau_d(q, \dot{q}) \in R^n$  are the input torque and the unknown disturbance related to the system signals, respectively.

The objective of our work is to make the output errors in (1) converge to a predefined neighborhood around zero with guaranteed transient performance in fixed time regardless of the initial conditions. Meanwhile, all the closed-loop signals can be bounded during the operation. The controller design strategy is demonstrated in Fig. 1.

### B. Preliminaries

*Property 1* (See [8]): The matrix  $M(q)$  in (1) is symmetric and positive definite. The term  $\dot{M}(q) - 2C(q, \dot{q})$  is skew symmetric and satisfies that  $\delta^T(\dot{M}(q) - 2C(q, \dot{q}))\delta = 0 \forall \delta \in R^n$ .

*Assumption 1*: The reference trajectory vector  $q_d \in R^n$  and its first-order derivative  $\dot{q}_d \in R^n$  are considered continuous and bounded.

*Lemma 1* (See [26]): For a general system  $\dot{x}(t) = g(x)$ , when the Lyapunov function  $V(x)$  satisfies the following inequality:  $\dot{V}(x) \leq -\mu_1 V^{\eta_1}(x) - \mu_2 V^{\eta_2}(x) + \rho$ , where  $\mu_1, \mu_2$ , and  $\rho$

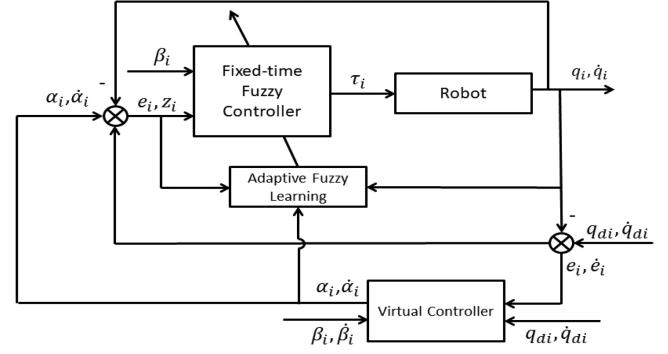


Fig. 1. Framework for the fixed-time fuzzy robot controller.

denote positive constants and the parameters  $\eta_1 \in (0, 1)$  and  $\eta_2 \in (1, \infty)$ , the system is practically fixed-time stable. And the residual set of the system can be represented as

$$\Omega_1 = \left\{ x \mid V \leq \min \left( [\rho / (\mu_1(1 - \theta))]^{1/\eta_1}, [\rho / (\mu_2(1 - \theta))]^{1/\eta_2} \right) \right\}$$

where  $\theta \in (0, 1)$ . And the convergence time of the system can be derived as  $T \leq T_{\max} = \frac{1}{\mu_1 \theta (1 - \eta_1)} + \frac{1}{\mu_2 \theta (\eta_2 - 1)}$ .

*Lemma 2* (See [27]): The following inequalities hold for any  $x_r \in R$  and  $y \in R^+$ :

$$\sum_{r=1}^n |x_r|^y \geq \left( \sum_{r=1}^n |x_r| \right)^y, \quad 0 < y \leq 1 \quad (2)$$

$$\sum_{r=1}^n |x_r|^y \geq n^{1-y} \left( \sum_{r=1}^n |x_r| \right)^y, \quad y > 1. \quad (3)$$

*Lemma 3* (See [28]): The following inequality holds for any  $a, b \in R$  and  $\epsilon \in R^+$ :

$$ab \leq \frac{\epsilon^{h_1}}{h_1} |a|^{h_1} + \frac{1}{h_2 \epsilon^{h_2}} |b|^{h_2} \quad (4)$$

where  $h_1, h_2 \in (1, \infty)$  are both positive constants, which satisfy that  $(h_1 - 1)(h_2 - 1) = 1$ .

*Fuzzy Logic System* (See [8]): As described in [8], an FLS is applied to approximate the unknown function  $f(Z) : R^k \rightarrow R$  in this article, where  $Z \in \Omega_C \subset R^k$  is the FLS input vector and  $k$  is the dimension of the FLS input. The fuzzy IF–THEN inference rules of the knowledge base in the FLS can be described as follows:

*Rule  $l$* : If  $Z_1$  is  $A_1^l$ ,  $Z_2$  is  $A_2^l, \dots, Z_k$  is  $A_k^l$ ,

then  $\psi$  is  $B^l$ ,  $l = 1, 2, \dots, N_0$

where  $Z = [Z_1, Z_2, \dots, Z_k]^T \in R^k$  and  $\psi$  are the input and output of the FLS, respectively.  $\mu_{A_j^l}(Z_j)$  and  $\mu_{B^l}(\psi)$  can be represented as the membership functions of the fuzzy sets  $A_j^l$  and  $B^l$ , respectively.  $N_0$  is the number of the fuzzy rules. Then, we represent the FLS as follows:

$$\psi = \frac{\sum_{l=1}^{N_0} \omega_l \prod_{j=1}^k \mu_{A_j^l}(Z_j)}{\sum_{l=1}^{N_0} \prod_{j=1}^k \mu_{A_j^l}(Z_j)} \quad (5)$$

where  $\omega_l$  is the point that achieves  $\mu_{B^l}(\omega_l) = 1$ . The membership function  $\mu_{A_j^l}(Z_j)$  is selected as a general Gaussian function

$$\mu_{A_j^l}(Z_j) = \exp \left[ -\frac{(Z_j - \sigma_j^l)^2}{\varpi_j^l} \right] \quad (6)$$

where  $\sigma_j^l$  and  $\varpi_j^l$  represent the center and the width, respectively. Define the fuzzy basic function as

$$S_l(Z) = \frac{\prod_{j=1}^k \mu_{A_j^l}(Z_j)}{\sum_{l=1}^{N_0} \prod_{j=1}^k \mu_{A_j^l}(Z_j)}. \quad (7)$$

Then, (5) can be rewritten as  $\psi = W^T S(Z)$ , where  $W = [\omega_1, \omega_2, \dots, \omega_{N_0}]^T$  and  $S(Z) = [S_1(Z), S_2(Z), \dots, S_{N_0}(Z)]^T$ , respectively.

**Lemma 4** (See [29]): For a continuous function  $f(Z)$  defined on a compact set  $\Omega_C$ , there exists an FLS such that

$$\sup_{Z \in \Omega_C} |f(Z) - W^T S(Z)| \leq \varsigma \quad (8)$$

for any given  $\varsigma > 0$ .

In terms of Lemma 4, the function  $f(Z)$  can be formulated as

$$f(Z) = W^{*T} S(Z) + \varepsilon(Z) \quad \forall Z \in \Omega_C \in R^k \quad (9)$$

where  $W^*, S(Z) \in R^{N_0}$  are the optimal weight vector and the regression vector, respectively. And  $|\varepsilon(Z)| \leq \bar{\varepsilon}$  denotes the approximation error of the FLS and is bounded by some positive constant  $\bar{\varepsilon}$  for all  $Z \in \Omega_C$ .

### III. CONTROL DESIGN

The tracking error signals are defined below before proceeding to the control design:

$$e = q - q_d, \quad z = \dot{q} - \alpha \quad (10)$$

where  $e = [e_1, e_2, \dots, e_n]^T \in R^n$  and  $z = [z_1, z_2, \dots, z_n]^T \in R^n$  are the position tracking error and the velocity tracking error vectors of the robot manipulator, respectively.  $q, q_d \in R^n$  denote the actual and desired trajectory, respectively. And  $\alpha = [\alpha_1, \alpha_2, \dots, \alpha_n]^T \in R^n$  is the virtual controller to be specified later.

Inspired by Chen et al. [30], a series of smoothly decreasing functions are defined as follows to constrain the tracking errors:

$$\beta_i = (\varrho_{0i} - \varrho_{\infty i})e^{-\vartheta_i t} + \varrho_{\infty i}, \quad i = 1, 2, \dots, n \quad (11)$$

where  $\varrho_{0i}, \varrho_{\infty i}, \vartheta_i$  ( $i = 1, 2, \dots, n$ ) are all properly designed positive constants. It can be noted that when the tracking error  $e_i$  is constrained in the range  $-\beta_i < e_i < \beta_i$ , the designed parameters  $\varrho_{0i}, \varrho_{\infty i}$ , and  $\vartheta_i$  can specify the tracking transient response of the system, i.e., the exponential term  $\vartheta_i$  regulates the convergence rate, and  $\varrho_{0i}$  and  $\varrho_{\infty i}$  can define the maximum overshoot and the steady error, respectively.

Inspired by Yang et al. [31], a coordinate transformation of the tracking error vector  $e$  is applied and a novel symmetric BLF is designed as follows:

$$\varphi = [\varphi_1, \varphi_2, \dots, \varphi_n]^T = \left[ \frac{e_1}{\beta_1}, \frac{e_2}{\beta_2}, \dots, \frac{e_n}{\beta_n} \right]^T \quad (12)$$

$$V_1 = \sum_{i=1}^n \frac{1}{2} \left( \frac{\varphi_i^2}{1 - \varphi_i^2} \right) \quad (13)$$

where  $\varphi_i$  is defined as the  $i$ th element of the vector  $\varphi \in R^n$  and the initial value of  $e_i$  satisfies that  $-\varrho_{0i} < e_i(0) < \varrho_{0i}$ .

Taking the derivative of (13) with respect to time combined with (12), we can obtain

$$\dot{V}_1 = \sum_{i=1}^n \frac{\varphi_i(\dot{e}_i \beta_i - e_i \dot{\beta}_i)}{\beta_i^2 (1 - \varphi_i^2)^2}. \quad (14)$$

From (10), we have

$$\dot{e}_i = \dot{q}_i - \dot{q}_{di} = z_i + \alpha_i - \dot{q}_{di}. \quad (15)$$

Substituting (15) into (14), we have

$$\dot{V}_1 = \sum_{i=1}^n \frac{\varphi_i z_i + \varphi_i \alpha_i - \varphi_i \dot{q}_{di} - \dot{\beta}_i \varphi_i^2}{\beta_i (1 - \varphi_i^2)^2}. \quad (16)$$

Then, the virtual controller  $\alpha_i$  is designed as follows:

$$\alpha_i = \dot{q}_{di} + \varphi_i \dot{\beta}_i - k_{1i} \beta_i \varphi_i^3 - k_{2i} \beta_i |\varphi_i|^{0.5} \text{sgn}(\varphi_i) (1 - \varphi_i^2)^{\frac{5}{4}} \quad (17)$$

where  $\text{sgn}(\cdot)$  represents the signum function and  $k_{1i}$  and  $k_{2i}$  are both positive constants.

Substituting (17) into (16), we have

$$\begin{aligned} \dot{V}_1 &= \sum_{i=1}^n \left[ \frac{\varphi_i z_i}{\beta_i (1 - \varphi_i^2)^2} - \frac{4k_{1i} \left(\frac{1}{2}\varphi_i^2\right)^2}{(1 - \varphi_i^2)^2} - \frac{k_{2i} |\varphi_i|^{\frac{3}{2}}}{(1 - \varphi_i^2)^{\frac{3}{4}}} \right] \\ &= \sum_{i=1}^n \left\{ \phi_i - 4k_{1i} \left[ \frac{\varphi_i^2}{2(1 - \varphi_i^2)} \right]^2 - \frac{k_{2i}}{\left(\frac{1}{2}\right)^{\frac{3}{4}}} \left[ \frac{\varphi_i^2}{2(1 - \varphi_i^2)} \right]^{\frac{3}{4}} \right\} \end{aligned} \quad (18)$$

where  $\phi_i = \frac{e_i z_i}{(\beta_i^2 - e_i^2)(1 - \varphi_i^2)}$ .

Then, a positive candidate Lyapunov function is defined as

$$V_2 = V_1 + \frac{1}{2} z^T M z \quad (19)$$

where  $M$  denotes the abbreviation of  $M(q)$  in (1).

Taking the derivative of (19) combined with (1) and (10), we can obtain

$$\dot{V}_2 = \dot{V}_1 + z^T (\tau - C\dot{q} - G - \tau_d - M\dot{\alpha}) + \frac{1}{2} z^T \dot{M} z \quad (20)$$

where  $C, G$ , and  $\tau_d$  are the abbreviation of  $C(q, \dot{q})$ ,  $G(q)$ , and  $\tau_d(q, \dot{q})$  in (1), respectively.

According to Property 1, we can rewrite (20) as

$$\dot{V}_2 = \dot{V}_1 + z^T (\tau - C\alpha - G - \tau_d - M\dot{\alpha}). \quad (21)$$

We define the unknown vector, which will be approximated by the FLSs as

$$F(Z) = -C\alpha - G - \tau_d - M\dot{\alpha} \quad (22)$$

where  $Z = [q^T, \dot{q}^T, \alpha^T, \dot{\alpha}^T]^T \in R^{4n}$  is the FLS input vector and  $F(Z) = [F_1(Z), F_2(Z), \dots, F_n(Z)]^T \in R^n$  is the  $n$ -dimensional unknown function vector.

Substituting (22) into (21), we can obtain

$$\dot{V}_2 = \dot{V}_1 + z^T [\tau + F(Z)]. \quad (23)$$

We exploit  $n$  FLSs in Section II-B to approximate the unknown functions  $F_i(Z)$  in (22) ( $i = 1, 2, \dots, n$ ). Then, we can represent the function  $F_i(Z)$  and its estimation  $\hat{F}_i(Z)$ , respectively, as

$$\begin{aligned} F_i(Z) &= W_i^{*T} S_i(Z) + \varepsilon_i \\ \hat{F}_i(Z) &= \hat{W}_i^T S_i(Z) \end{aligned} \quad (24)$$

where  $W_i^{*T}, \hat{W}_i^T \in R^{N_{0i}}$  are the optimal and estimated FLS weight vector, respectively.  $N_{0i}$  is the fuzzy rules number of the  $i$ th FLS.  $S_i(Z) \in R^{N_{0i}}$  is the  $i$ th FLS's basis function vector. According to the definition of the FLS, there is a positive constant  $\bar{\varepsilon}_i$  such that the estimation error of the FLS in (24) is bounded by  $\varepsilon_i \leq \bar{\varepsilon}_i$ .

Then, the adaptive fixed-time fuzzy control law for the system (1) is designed as

$$\begin{aligned} \tau_i &= -\frac{e_i}{(\beta_i^2 - e_i^2)(1 - \varphi_i^2)} - k_{3i} z_i^3 - \hat{F}_i \\ &\quad - k_{4i} |z_i|^{0.5} \tanh(z_i) - \frac{3}{4} z_i \end{aligned} \quad (25)$$

where  $k_{3i}$  and  $k_{4i}$  are all proper positive constants and  $\hat{F}_i$  is the abbreviation of  $\hat{F}_i(Z)$  in (24).

A novel FLS weight adaptive law is designed as

$$\dot{\hat{W}}_i = Q_i \left[ z_i S_i - \gamma_{1i} \hat{W}_i - \gamma_{2i} (\hat{W}_i^T \hat{W}_i) \hat{W}_i \right] \quad (26)$$

where the term  $Q_i$  is a positive-definite diagonal matrix. The parameters  $\gamma_{1i}$  and  $\gamma_{2i}$  are both designed positive constants.  $S_i$  denotes the abbreviation of  $S_i(Z)$  in (24).

**Theorem 1:** Consider the uncertain robot system in (1) with the initial conditions  $e_i(0)$  satisfying that  $-\beta_i(0) < e_i(0) < \beta_i(0)$ , when the virtual controllers  $\alpha_i$  in (17), the control law  $\tau_i$  in (25), and the adaptation law in (26) are employed; the tracking errors  $e_i$  can achieve fixed-time convergence without violating the symmetric output constraints  $-\beta_i < e_i < \beta_i$ . Meanwhile, all the closed-loop signals in the system can be proved to be bounded.

*Proof:* The global system Lyapunov function is designed as follows:

$$V = V_2 + \frac{1}{2} \sum_{i=1}^n \tilde{W}_i^T Q_i^{-1} \tilde{W}_i \quad (27)$$

where  $\tilde{W}_i = W_i^* - \hat{W}_i$  denotes the estimation vector of the optimal FLS weight  $W_i^*$ .

Substituting (18) and (23)–(26) into the derivative of  $V$ , we can obtain

$$\begin{aligned} \dot{V} &= \sum_{i=1}^n \left\{ -4k_{1i} \left[ \frac{\varphi_i^2}{2(1 - \varphi_i^2)} \right]^2 - \frac{k_{2i}}{\left(\frac{1}{2}\right)^{\frac{3}{4}}} \left[ \frac{\varphi_i^2}{2(1 - \varphi_i^2)} \right]^{\frac{3}{4}} \right. \\ &\quad - 4k_{3i} \left( \frac{1}{2} z_i^2 \right)^2 - \frac{k_{4i}}{\left(\frac{1}{2}\right)^{\frac{3}{4}}} \times \left( \frac{1}{2} \right)^{\frac{3}{4}} z_i |z_i|^{\frac{1}{2}} \tanh(z_i) \\ &\quad - z_i \hat{W}_i^T S_i + z_i W_i^{*T} S_i + z_i \varepsilon_i - \frac{3}{4} z_i^2 - z_i \tilde{W}_i^T S_i \\ &\quad \left. + \gamma_{1i} \tilde{W}_i^T \hat{W}_i + \gamma_{2i} (\hat{W}_i^T \hat{W}_i) \tilde{W}_i^T \hat{W}_i \right\}. \end{aligned} \quad (28)$$

According to [31], for any  $\nu \in R$  and  $c > 0$ , the following inequality can be held:

$$0 \leq |\nu| - \nu \tanh\left(\frac{\nu}{c}\right) \leq \iota c \quad (29)$$

where  $\iota$  is a positive constant satisfying that  $\iota = e^{-(\iota+1)}$ , i.e.,  $\iota = 0.2785$ .

Then, the following inequality can be obtained:

$$-z_i \tanh(z_i) \leq -|z_i| + 0.2785 \quad (30)$$

where  $c \in R^+$  in (29) has been selected as  $c = 1$ .

With Young's inequality being employed, we can obtain the following inequality:

$$0.2785 k_{4i} |z_i|^{\frac{1}{2}} \leq \frac{3}{4} (0.2785 k_{4i})^{\frac{4}{3}} + \frac{1}{4} z_i^2. \quad (31)$$

By (30) and (31), we can conclude that

$$\begin{aligned} & -\frac{k_{4i}}{\left(\frac{1}{2}\right)^{\frac{3}{4}}} \times \left(\frac{1}{2}\right)^{\frac{3}{4}} z_i |z_i|^{\frac{1}{2}} \tanh(z_i) \\ & \leq \frac{k_{4i}}{\left(\frac{1}{2}\right)^{\frac{3}{4}}} \times \left(\frac{1}{2}\right)^{\frac{3}{4}} |z_i|^{\frac{1}{2}} (-|z_i| + 0.2785) \\ & \leq -\frac{k_{4i}}{\left(\frac{1}{2}\right)^{\frac{3}{4}}} \left(\frac{1}{2} z_i^2\right)^{\frac{3}{4}} + 0.2785 k_{4i} |z_i|^{\frac{1}{2}} \\ & \leq -\frac{k_{4i}}{\left(\frac{1}{2}\right)^{\frac{3}{4}}} \left(\frac{1}{2} z_i^2\right)^{\frac{3}{4}} + \frac{1}{4} z_i^2 + \zeta_{1i} \end{aligned} \quad (32)$$

where  $\zeta_{1i} = \frac{3}{4} (0.2785 k_{4i})^{\frac{4}{3}}$ .

For the term  $z_i \varepsilon_i$  in (28), the following inequality can be obtained:

$$z_i \varepsilon_i \leq |z_i| |\varepsilon_i| \leq \frac{1}{2} z_i^2 + \frac{1}{2} \varepsilon_i^2 \leq \frac{1}{2} z_i^2 + \frac{1}{2} \bar{\varepsilon}_i^2. \quad (33)$$

Substituting (32) and (33) into (28), we can obtain

$$\begin{aligned} \dot{V} &= \sum_{i=1}^n \left\{ -4k_{1i} \left[ \frac{\varphi_i^2}{2(1 - \varphi_i^2)} \right]^2 - \frac{k_{2i}}{\left(\frac{1}{2}\right)^{\frac{3}{4}}} \left[ \frac{\varphi_i^2}{2(1 - \varphi_i^2)} \right]^{\frac{3}{4}} \right. \\ &\quad - 4k_{3i} \left( \frac{1}{2} z_i^2 \right)^2 - \frac{k_{4i}}{\left(\frac{1}{2}\right)^{\frac{3}{4}}} \left( \frac{1}{2} z_i^2 \right)^{\frac{3}{4}} + \frac{1}{4} z_i^2 + \zeta_{1i} + \frac{1}{2} z_i^2 \\ &\quad \left. + \frac{1}{2} \bar{\varepsilon}_i^2 - \frac{3}{4} z_i^2 + \gamma_{1i} \tilde{W}_i^T \hat{W}_i + \gamma_{2i} (\hat{W}_i^T \hat{W}_i) \tilde{W}_i^T \hat{W}_i \right\}. \end{aligned} \quad (34)$$

Utilizing Young's inequality, we have

$$\begin{aligned}\gamma_{1i} \tilde{W}_i^T \hat{W}_i &= \gamma_{1i} \tilde{W}_i^T (W_i^* - \tilde{W}_i) \\ &\leq -\gamma_{1i} \|\tilde{W}_i\|^2 + \gamma_{1i} \left( \frac{\|\tilde{W}_i\|^2}{2} + \frac{\|W_i^*\|^2}{2} \right) \\ &\leq \gamma_{1i} \left( \frac{\|W_i^*\|^2}{2} - \frac{\|\tilde{W}_i\|^2}{2} \right).\end{aligned}\quad (35)$$

Then, for any given positive constant  $\gamma_{3i} > 0$ , the following formulas can be concluded:

$$\begin{aligned}& -\frac{\gamma_{1i}}{2} \|\tilde{W}_i\|^2 \\ &= -\frac{1}{4} \left[ \|\tilde{W}_i\| \sqrt{\gamma_{1i}} - (2)^{\frac{1}{4}} \frac{\gamma_{3i} \sqrt{\|\tilde{W}_i\|}}{\sqrt{\gamma_{1i}}} \right]^2 - \gamma_{3i} \left( \frac{\|\tilde{W}_i\|^2}{2} \right)^{\frac{3}{4}} \\ & \quad - \frac{\gamma_{1i}}{4} \|\tilde{W}_i\|^2 + \frac{\sqrt{2}}{4} \times \frac{\gamma_{3i}^2}{\gamma_{1i}} \|\tilde{W}_i\|\end{aligned}\quad (36)$$

and

$$\begin{aligned}& \frac{\sqrt{2}}{4} \times \frac{\gamma_{3i}^2}{\gamma_{1i}} \|\tilde{W}_i\| \\ &= \frac{1}{\gamma_{1i}} \left( \frac{\gamma_{1i} \|\tilde{W}_i\|}{2} \times \frac{\sqrt{2} \gamma_{3i}^2}{2 \gamma_{1i}} \right) \\ &\leq \frac{1}{\gamma_{1i}} \left( \frac{\gamma_{1i}^2 \|\tilde{W}_i\|^2}{8} + \frac{\gamma_{3i}^4}{4 \gamma_{1i}^2} \right) \\ &\leq \frac{\gamma_{1i}}{8} \|\tilde{W}_i\|^2 + \frac{\gamma_{3i}^4}{4 \gamma_{1i}^3}.\end{aligned}\quad (37)$$

Substituting (36) and (37) into (35), we can obtain

$$\gamma_{1i} \tilde{W}_i^T \hat{W}_i \leq -\gamma_{3i} \left( \frac{\|\tilde{W}_i\|^2}{2} \right)^{\frac{3}{4}} + \frac{\gamma_{1i}}{2} \|W_i^*\|^2 + \frac{\gamma_{3i}^4}{4 \gamma_{1i}^3}. \quad (38)$$

In addition, we have

$$\begin{aligned}& \gamma_{2i} (\hat{W}_i^T \hat{W}_i) \tilde{W}_i^T \hat{W}_i \\ &= \gamma_{2i} \left[ (W_i^* - \tilde{W}_i)^T (W_i^* - \tilde{W}_i) \right] \tilde{W}_i^T (W_i^* - \tilde{W}_i) \\ &= \gamma_{2i} \left[ \|W_i^*\|^2 \tilde{W}_i^T W_i^* - \|W_i^*\|^2 \|\tilde{W}_i\|^2 - 2 (\tilde{W}_i^T W_i^*)^2 \right. \\ & \quad \left. + 3 \|\tilde{W}_i\|^2 \tilde{W}_i^T W_i^* - \|\tilde{W}_i\|^4 \right].\end{aligned}\quad (39)$$

From Lemma 3, we can conclude that

$$\|W_i^*\|^2 \tilde{W}_i^T W_i^* \leq \|W_i^*\|^3 \|\tilde{W}_i\| \leq \|W_i^*\|^2 \|\tilde{W}_i\|^2$$

$$+ \frac{1}{4} \|W_i^*\|^4 \quad (40)$$

$$\begin{aligned}& 3 \|\tilde{W}_i\|^2 \tilde{W}_i^T W_i^* \\ &\leq 3 \|\tilde{W}_i\|^3 \|W_i^*\| \\ &\leq 3 \left[ \frac{3 \xi_i^{\frac{4}{3}}}{4} \|\tilde{W}_i\|^4 + \frac{1}{4 \xi_i^4} \|W_i^*\|^4 \right]\end{aligned}\quad (41)$$

where  $0 < \xi_i < (\frac{2}{3})^{\frac{3}{2}}$  is a positive parameter specified by the designer.

*Remark 1:* Two auxiliary positive parameters  $\gamma_{3i}$  and  $\xi_i$  are introduced to demonstrate the fixed-time convergence for the system, but they are not involved in the control scheme design. The control performance can be optimized by appropriately selecting the control parameters in (17), (25), and (26).

Substituting (38)–(41) into (34), we can obtain

$$\begin{aligned}\dot{V} &\leq \sum_{i=1}^n \left\{ -4k_{1i} \left[ \frac{\varphi_i^2}{2(1-\varphi_i^2)} \right]^2 - \frac{k_{2i}}{(\frac{1}{2})^{\frac{3}{4}}} \left[ \frac{\varphi_i^2}{2(1-\varphi_i^2)} \right]^{\frac{3}{4}} \right. \\ & \quad - 4k_{3i} \left( \frac{1}{2} z_i^2 \right)^2 - \frac{k_{4i}}{(\frac{1}{2})^{\frac{3}{4}}} \left( \frac{1}{2} z_i^2 \right)^{\frac{3}{4}} + \zeta_{1i} + \frac{1}{2} \bar{\varepsilon}_i^2 \\ & \quad + \frac{\gamma_{1i}}{2} \|W_i^*\|^2 + \frac{\gamma_{3i}^4}{4 \gamma_{1i}^3} - \gamma_{3i} \left( \frac{\|\tilde{W}_i\|^2}{2} \right)^{\frac{3}{4}} \\ & \quad + \gamma_{2i} \left[ \frac{1}{4} \|W_i^*\|^4 - 2 (\tilde{W}_i^T W_i^*)^2 + \frac{3}{4 \xi_i^4} \|W_i^*\|^4 \right. \\ & \quad \left. \left. + \left( \frac{9 \xi_i^{\frac{4}{3}}}{4} - 1 \right) \|\tilde{W}_i\|^4 \right] \right\} \\ &\leq \sum_{i=1}^n \left\{ -4k_{1i} \left[ \frac{\varphi_i^2}{2(1-\varphi_i^2)} \right]^2 - \frac{k_{2i}}{(\frac{1}{2})^{\frac{3}{4}}} \left[ \frac{\varphi_i^2}{2(1-\varphi_i^2)} \right]^{\frac{3}{4}} \right. \\ & \quad - 4k_{3i} \left( \frac{1}{2} z_i^2 \right)^2 - \frac{k_{4i}}{(\frac{1}{2})^{\frac{3}{4}}} \left( \frac{1}{2} z_i^2 \right)^{\frac{3}{4}} + \zeta_{2i} \\ & \quad \left. \gamma_{3i} \left( \frac{\|\tilde{W}_i\|^2}{2} \right)^{\frac{3}{4}} - \gamma_{2i} \left( 4 - 9 \xi_i^{\frac{4}{3}} \right) \left( \frac{1}{2} \|\tilde{W}_i\|^2 \right)^2 \right\}\end{aligned}\quad (42)$$

where  $\zeta_{2i} = \zeta_{1i} + \frac{1}{2} \bar{\varepsilon}_i^2 + \frac{\gamma_{1i}}{2} \|W_i^*\|^2 + \frac{\gamma_{3i}^4}{4 \gamma_{1i}^3} + \gamma_{2i} \left( \frac{1}{4} + \frac{3}{4 \xi_i^4} \right) \|W_i^*\|^4$  is a positive constant.

Combined with Lemma 2, we can conclude that

$$\dot{V} \leq \sum_{i=1}^n -4k_{1i} \left[ \frac{\varphi_i^2}{2(1-\varphi_i^2)} \right]^2 - \sum_{i=1}^n \frac{k_{2i}}{(\frac{1}{2})^{\frac{3}{4}}} \left[ \frac{\varphi_i^2}{2(1-\varphi_i^2)} \right]^{\frac{3}{4}}$$

$$\begin{aligned}
& - \min \left[ \frac{4k_{3i}}{\lambda_{\max}^2(M)n} \right] \left( \frac{1}{2} z^T M z \right)^2 \\
& - \min \left[ \frac{\gamma_{2i} \left( 4 - 9\xi_i^{\frac{4}{3}} \right)}{\lambda_{\max}^2(Q_i^{-1})n} \right] \left( \sum_{i=1}^n \frac{1}{2} \tilde{W}_i^T Q_i^{-1} \tilde{W}_i \right)^2 \\
& - \min \left[ \frac{k_{4i}}{\left( \frac{1}{2} \right)^{\frac{3}{4}} \lambda_{\max}^{\frac{3}{4}}(M)} \right] \left( \frac{1}{2} z^T M z \right)^{\frac{3}{4}} \\
& - \min \left[ \frac{\gamma_{3i}}{\lambda_{\max}^{\frac{3}{4}}(Q_i^{-1})} \right] \left( \sum_{i=1}^n \frac{1}{2} \tilde{W}_i^T Q_i^{-1} \tilde{W}_i \right)^{\frac{3}{4}} + \zeta_2 \\
& \leq -P_1 V^2 - P_2 V^{\frac{3}{4}} + \zeta_2 \quad (43)
\end{aligned}$$

$$\text{where } P_1 = \frac{1}{3n} \min \left[ 4k_{1i}, \frac{4k_{3i}}{\lambda_{\max}^2(M)}, \frac{\gamma_{2i}(4-9\xi_i^{\frac{4}{3}})}{\lambda_{\max}^2(Q_i^{-1})} \right], P_2 = \min \left[ \frac{k_{2i}}{\left( \frac{1}{2} \right)^{\frac{3}{4}}}, \frac{k_{4i}}{\left( \frac{1}{2} \right)^{\frac{3}{4}} \lambda_{\max}^{\frac{3}{4}}(M)}, \frac{\gamma_{3i}}{\lambda_{\max}^{\frac{3}{4}}(Q_i^{-1})} \right], \text{ and } \zeta_2 = \sum_{i=1}^n \left[ \zeta_{1i} + \frac{1}{2} \bar{e}_i^2 + \frac{\gamma_{1i}}{2} \|W_i^*\|^2 + \frac{\gamma_{3i}}{4\gamma_{1i}^{\frac{3}{4}}} + \gamma_{2i} \left( \frac{1}{4} + \frac{3}{4\xi_i^{\frac{4}{3}}} \right) \|W_i^*\|^4 \right].$$

Then, the boundedness of  $V$  is demonstrated in two stages:  $V \in [1, +\infty)$  and  $V \in [0, 1)$ . We have the following inequalities:

$$\dot{V} \leq -P_1 V^2 - P_2 V^{\frac{3}{4}} + \zeta_2 \leq -P_1 V + \zeta_2, \quad V \geq 1 \quad (44)$$

$$\dot{V} \leq -P_1 V^2 - P_2 V^{\frac{3}{4}} + \zeta_2 \leq -P_2 V + \zeta_2, \quad 0 \leq V < 1. \quad (45)$$

According to the definition of the Lyapunov function in (13), (19), and (27), the initial condition of  $-\varrho_{0i} < e_i(0) < \varrho_{0i}$ , and the results in [31] and [32], we can infer the boundedness of  $\frac{\varphi_i^2}{1-\varphi_i^2}$ ,  $z_i$ , and  $\tilde{W}_i$ , respectively. Then, the transient performance  $-\beta_i < e_i < \beta_i$  can be achieved during the operation from (12). From the definition of  $q = q_d + e$  with Assumption 1, the boundedness of  $q$  can be inferred. Then, combined with the boundedness of  $\beta_i$  and  $\dot{\beta}_i$ , we can conclude the boundedness of the virtual controller  $\alpha_i$ . Then,  $\dot{q}_i = z_i + \alpha_i$  is also bounded. According to the boundedness of  $\hat{W}_i = W_i^* - \tilde{W}_i$  and the Gaussian basis function vector  $S_i(Z)$ , the controller in (25) is also bounded. Therefore, all the signals in the closed-loop system have been proved to be bounded.

In addition, from (43) and Lemma 1, we can conclude that the system is fixed-time convergent regardless of the initial conditions. And the residual set of the system can be shown as

$$\Omega_g = \left\{ x | V \leq \min \left[ \left( \frac{\zeta_2}{P_2(1-\theta_1)} \right)^{\frac{4}{3}}, \left( \frac{\zeta_2}{P_1(1-\theta_1)} \right)^{\frac{1}{2}} \right] \right\} \quad (46)$$

where  $\theta_1$  is a positive constant satisfying  $\theta_1 \in (0, 1)$ .

And the convergence time can be expressed as

$$T_{g\max} = \frac{4}{P_2\theta_1} + \frac{1}{P_1\theta_1}. \quad (47)$$

So far, the proof of Theorem 1 has been completed.

*Remark 2:* In the recent fixed-time control research [23]–[25], the weight estimation  $\|\tilde{W}_i\|$  is assumed to be bounded in the fixed-time stability analysis, which lacks theoretical basis. In our work, a novel adaptive law is proposed such that the boundedness of  $\|\tilde{W}_i\|$  can be proved theoretically instead of being assumed. Moreover, the system output  $e_i$  can be symmetrically constrained by  $(-\beta_i, \beta_i)$  due to the introduction of the BLF in (13) such that the desired transient performance can be guaranteed. Therefore, the superiority of our work can be shown compared with the existing control schemes.

*Remark 3:* From (47) and the definitions of  $P_1$  and  $P_2$ , we can conclude that larger  $k_{1i}$ ,  $k_{2i}$ ,  $k_{3i}$ ,  $k_{4i}$ ,  $Q_i$ ,  $\gamma_{2i}$ , and  $\gamma_{3i}$  will make larger  $P_1$  and  $P_2$ ; thus, faster convergence can be achieved. In addition, by appropriately selecting the parameters  $\varrho_{0i}$ ,  $\varrho_{\infty i}$ , and  $\vartheta_i$  in (11), the tracking errors can be constrained in the predefined range  $(-\beta_i, \beta_i)$  rather than the uncertain residual set in (46) because of the introduction of the BLF in (13). It should be noted that larger control gains  $k_{1i}$ ,  $k_{2i}$ ,  $k_{3i}$ , and  $k_{4i}$  and smaller steady-state error  $\varrho_{\infty i}$  will lead to larger control torque, which may practically exceed the input upper limit of the robot in accordance with the expression in (25). Therefore, in the practical application, the parameters can be selected empirically combined with the control performance and the mechanical limitation.

#### IV. SIMULATION STUDY

In this section, a two-DOF robot model is selected to illustrate the effectiveness and superiority of the fixed-time fuzzy controller in (25).

##### A. Simulation Setup

When  $n = 2$  in (1),  $M(q)$ ,  $C(q, \dot{q})$ ,  $G(q)$ , and  $\tau_d(q, \dot{q})$  of the robot system are defined as follows:

$$\begin{aligned}
M(q) &= \begin{bmatrix} M_{11} & M_{12} \\ M_{21} & M_{22} \end{bmatrix} \\
C(q, \dot{q}) &= \begin{bmatrix} C_{11} & C_{12} \\ C_{21} & C_{22} \end{bmatrix} \\
G(q) &= \begin{bmatrix} G_1 \\ G_2 \end{bmatrix} \\
\tau_d(q, \dot{q}) &= \begin{bmatrix} \tau_{d1} \\ \tau_{d2} \end{bmatrix} \quad (48)
\end{aligned}$$

where

$$M_{11} = m_1 l_{c1}^2 + m_2 (l_{c1}^2 + l_{c2}^2 + 2l_1 l_{c2} \cos q_2) + I_1 + I_2$$

$$M_{12} = M_{21} = m_2 (l_{c2}^2 + l_1 l_{c2} \cos q_2) + I_2$$

$$M_{22} = m_2 l_{c2}^2 + I_2$$

$$C_{11} = -m_2 l_1 l_{c2} \dot{q}_2 \sin q_2$$

$$C_{12} = -m_2 l_1 l_{c2} (\dot{q}_1 + \dot{q}_2) \sin q_2$$

$$C_{21} = m_2 l_1 l_{c2} \dot{q}_1 \sin q_2$$

$$\begin{aligned}
C_{22} &= 0 \\
G_1 &= (m_1 l_{c2} + m_2 l_1) g \cos q_1 + m_2 l_{c2} g \cos(q_1 + q_2) \\
G_2 &= m_2 l_{c2} g \cos(q_1 + q_2) \\
\tau_{d1} &= \sin(0.5q_1) + \sin(0.3\dot{q}_1) \\
\tau_{d2} &= \sin(0.5q_2) + \sin(0.3\dot{q}_2)
\end{aligned} \quad (49)$$

with  $m_1 = m_2 = 2 \text{ kg}$  and  $l_1 = l_2 = 0.5 \text{ m}$  being the  $i$ th ( $i = 1, 2$ ) link's mass and length, respectively.  $g$  and  $l_{ci}$  ( $i = 1, 2$ ) denote the gravitational acceleration and the distance between the  $(i - 1)$ th joint and the mass center of the  $i$ th link, respectively. The inertia of the  $i$ th link can be calculated as  $I_1 = I_2 = 0.125 \text{ kg} \cdot \text{m}^2$ .

The desired trajectory of the two joints is set to be  $q_d = [\pi/2 + 0.7\cos(2t) - 0.7, \pi/4 + 0.8\sin(2t)]^T$ . In order to show the fixed-time convergence of the system regardless of the initial conditions, the initial errors of the two links are set as  $[0.1, -0.1]^T$ ,  $[0.2, -0.2]^T$ ,  $[0.4, -0.4]^T$ , and  $[0.8, -0.8]^T$  respectively. Moreover, the transient performance functions  $\beta_1$  and  $\beta_2$  are set as

$$\beta_1 = \beta_2 = (1 - 0.01)e^{-2t} + 0.01. \quad (50)$$

The parameters in the controller design can be set as  $k_{11} = k_{21} = 4$ ,  $k_{12} = k_{22} = 1.8$ ,  $k_{31} = k_{41} = 10$ , and  $k_{32} = k_{42} = 5$ . And the parameters of the FLS adaptive law in (26) are set as  $Q_1 = \text{diag}\{1\}$ ,  $Q_2 = \text{diag}\{0.7\}$ ,  $\gamma_{11} = \gamma_{21} = 0.02$ , and  $\gamma_{12} = \gamma_{22} = 0.008$ .

An adaptive NN controller in [33] has been applied to the same two-DOF robot manipulator to further show the superiority of our control scheme. With the two different controllers, the physical parameters of the system, the desired trajectory, and the output constraints are all set the same as that in the above simulation study, and the initial position error is set the same as  $[0.1, -0.1]^T$ . By the repeated simulation to achieve better control performance, for the virtual controller in (51), the NN controller in (52), and the adaptive law in (53) in [33]

$$\alpha_{1i} = \dot{y}_{di} - k_{1i}e_{1i} \quad (51)$$

$$u_i = -k_{2i}e_{2i} - \frac{1}{2a_i^2}e_{2i}\hat{\theta}_i\phi^T\phi \quad (52)$$

$$\dot{\hat{\theta}}_i = \frac{\lambda_i}{2a_i^2}e_{2i}\phi^T\phi - \gamma_i\hat{\theta}_i \quad (53)$$

the parameters are, respectively, set as  $k_1 = [40, 25]^T$ ,  $k_2 = [19.6, 9.6]^T$ ,  $a_i = 1.5$ , and  $\gamma_i = 1$  for  $i = 1, 2$ .

### B. Simulation Results

The simulation results are shown in Figs. 3–8. With four different initial conditions, we can see that the tracking error  $e_i$  and the FLS weight estimation  $\hat{W}_i$  can all achieve fixed-time convergence from Figs. 3, 4, and 6. Meanwhile, the tracking error  $e_i$  can be constrained in the range  $-\beta_i < e_i < \beta_i$  during

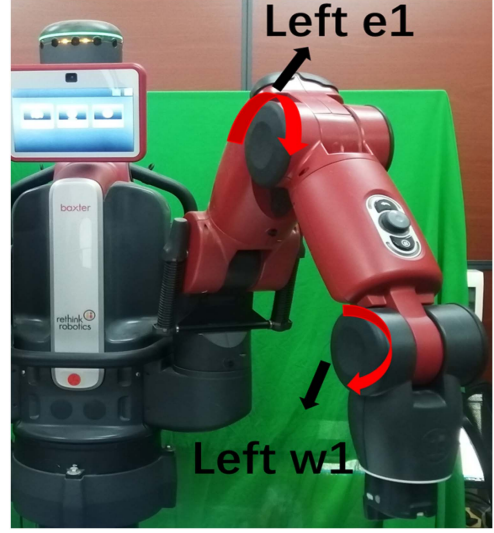


Fig. 2. Schematic diagram of the Baxter robot for the experiment.

the operation; thus, the predefined transient performance can be guaranteed. The bounded control input  $\tau_i$  with different initial conditions is shown in Fig. 5. Then, all the closed-loop system signals are all bounded from all the above results.

The comparison simulation results are shown in Figs. 7 and 8, from which we can see that both the controllers can achieve quite good tracking performance. In fact, better tracking performance can be achieved when more proper parameters are selected in [33] during our massive repeated simulation. However, when the selected parameters in [33] are suboptimal, the predefined output constraints may be violated, such as the results in Fig. 7. However, the same predefined transient constraints can always be guaranteed with the proposed control scheme. Thus, the superiority of our work has been shown in terms of the transient performance. We can see that the tracking performance of the proposed method is worse than that of the comparison method for a short period of time from the initial moment in the simulation, which may be because the initial torque of the proposed controller is relatively large; thus, a relatively oscillatory response has been caused. In addition, the weight estimation can always achieve faster convergence with the proposed control scheme in our massive repeated comparison simulation, such as the results in Fig. 8, which has shown the superiority of the proposed scheme from another level.

### V. EXPERIMENTAL STUDY

The Baxter bimanual robot, as shown in Fig. 2, is used to further demonstrate the practicability of the proposed controller. It is equipped with two seven-DOF arms whose upper limit of the control torques applying to the joints is  $50 \text{ N} \cdot \text{m}$  (the first four joints) and  $15 \text{ N} \cdot \text{m}$  (the last three joints) for each arm. For simplicity and without loss of generality, we choose the fourth and sixth joints of the left arm, i.e., *left e1* and *left w1* for the experiment, which has been marked out in the diagram.

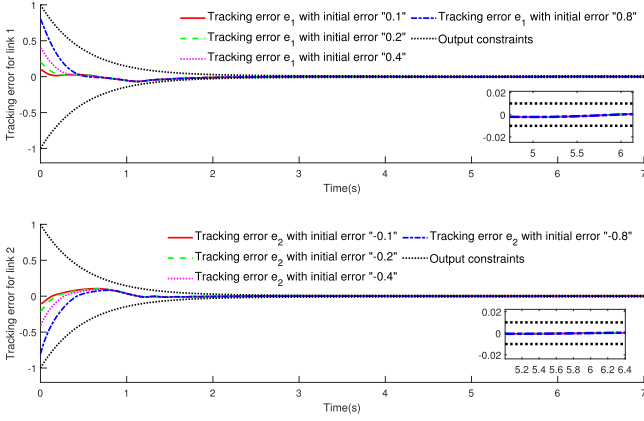


Fig. 3. Tracking error of the two-DOF robot manipulator with the proposed controller.

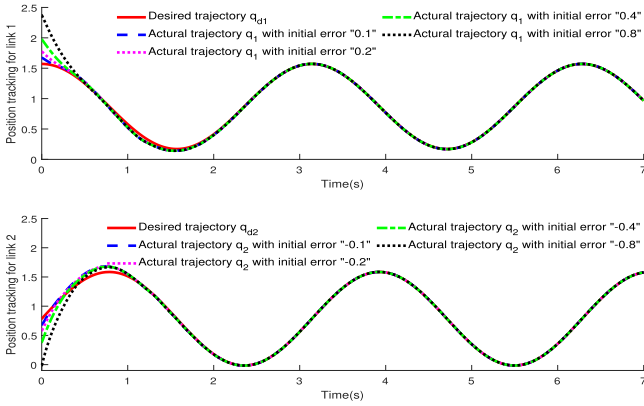


Fig. 4. Trajectory tracking of the two-DOF robot manipulator with the proposed controller.

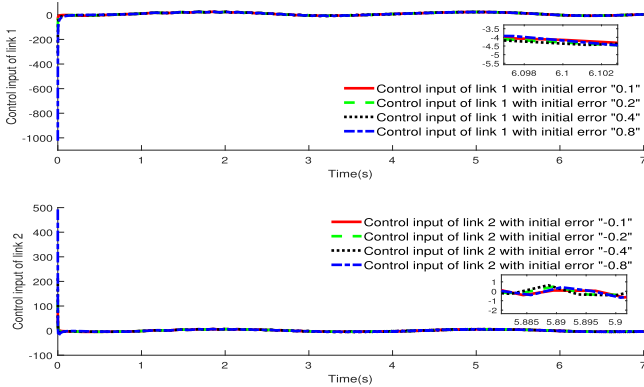


Fig. 5. Control input of the two-DOF robot manipulator with the proposed controller.

#### A. Experimental Setup

The desired trajectory is set to  $q_d = [\pi/2 + 0.6\cos(2t) - 0.6, \pi/4 + 0.8\sin(2t)]^T$ . In order to show the fixed-time convergence of the system regardless of the initial conditions, the initial errors of the two joints are set as  $[0.1, -0.1]^T$ ,  $[0.2, -0.2]^T$ ,

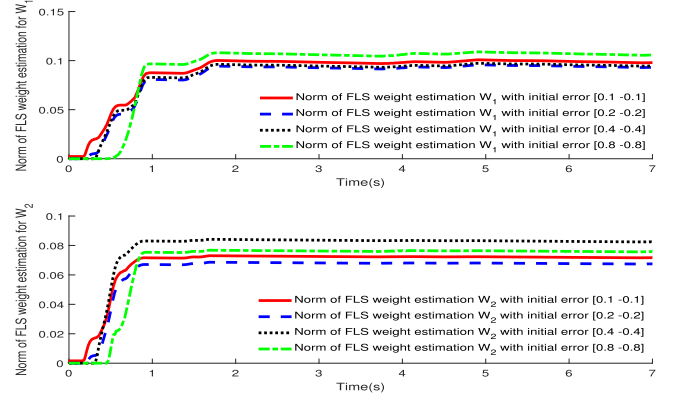


Fig. 6. FLS weight estimation of the two-DOF robot manipulator with the proposed controller.

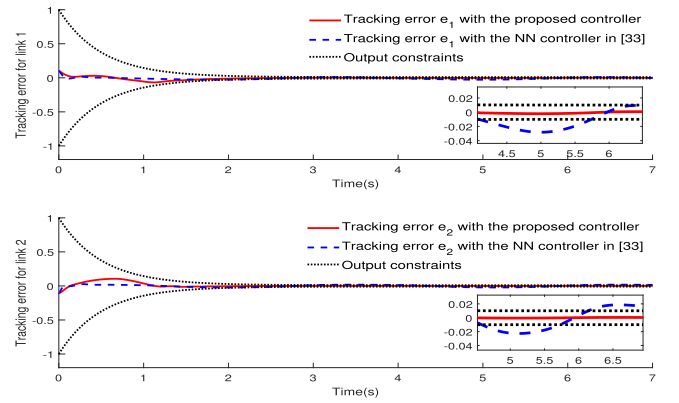


Fig. 7. Tracking error of the two-DOF robot manipulator with the proposed controller and the controller in [33].

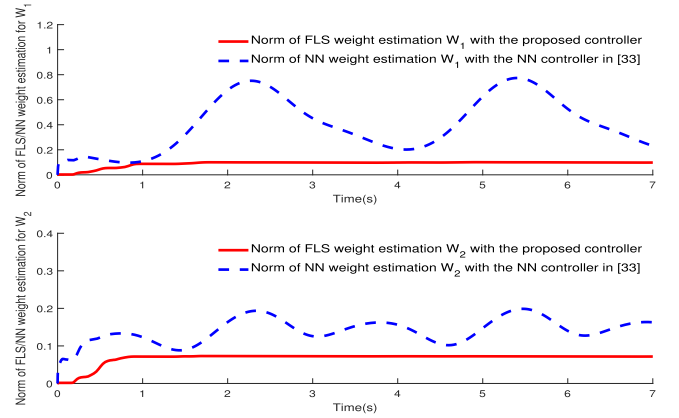


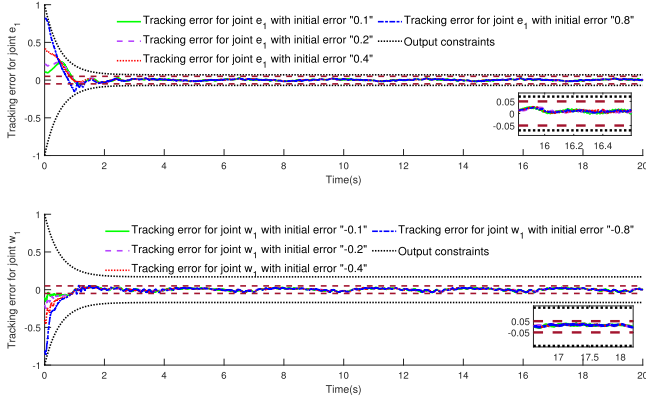
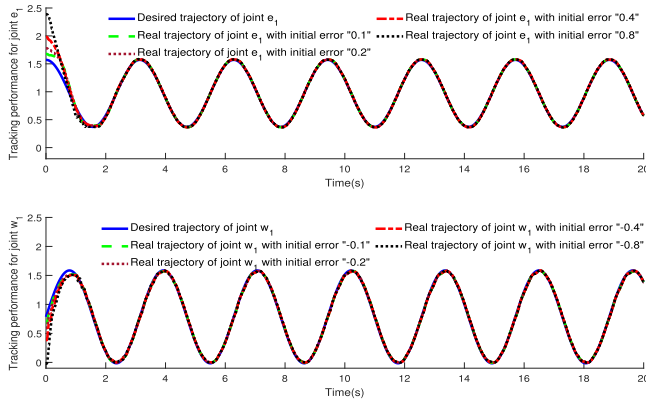
Fig. 8. FLS/NN weight estimation of the two-DOF robot manipulator with the proposed controller and the controller [33].

$[0.4, -0.4]^T$ , and  $[0.8, -0.8]^T$ , respectively. Moreover, the transient performance functions  $\beta_1$  and  $\beta_2$  are, respectively, set as

$$\beta_1 = (1 - 0.07)e^{-2t} + 0.07 \quad (54)$$

$$\beta_2 = (1 - 0.17)e^{-2t} + 0.17. \quad (55)$$

The parameters in the controller design can be set as  $k_{11} = 1, k_{21} = 1, k_{12} = 3.5, k_{22} = 1, k_{31} = 1, k_{41} = 1, k_{32} = 0.3$ , and  $k_{42} = 2$ . And the parameters of the FLS adaptive

Fig. 9. Tracking error of left  $e_1$  and  $w_1$  joints with the proposed controller.Fig. 10. Trajectory tracking of left  $e_1$  and  $w_1$  joints with the proposed controller.

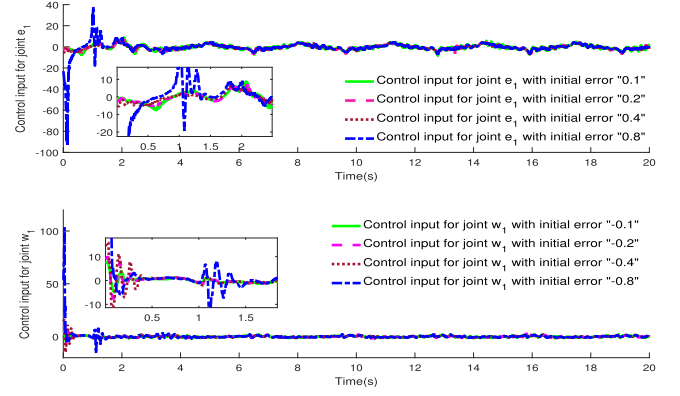
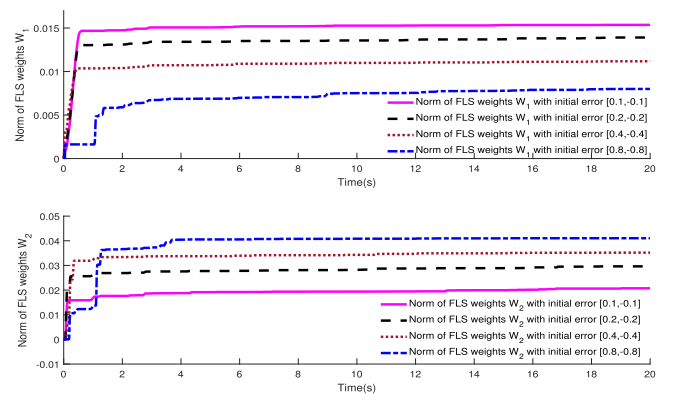
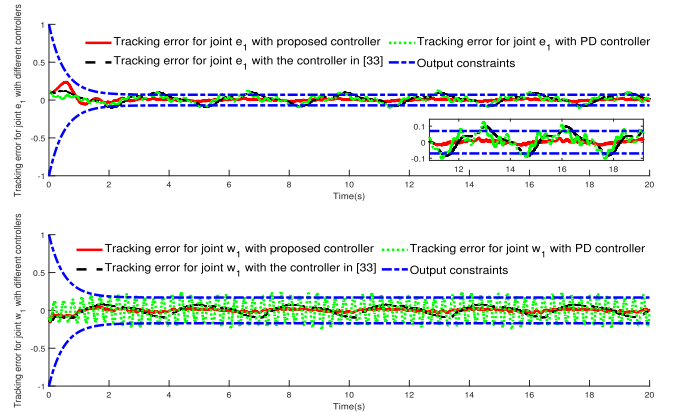
law in (26) are set as  $Q_1 = \text{diag}\{0.2\}$ ,  $Q_2 = \text{diag}\{0.8\}$ ,  $\gamma_{11} = \gamma_{21} = 15$ , and  $\gamma_{12} = \gamma_{22} = 10$ .

**Remark 4:** From (25), we can conclude that when  $e_i$  are close to the limits of the output constraints  $\pm\beta_i$ , very large control torque will be made; then, the input upper limit of the Baxter joints may be violated. Therefore, the slightly larger  $\varrho_{\infty 1}$  and  $\varrho_{\infty 2}$  are designed in (54) and (55).

An adaptive NN controller in [33] and a proportional-derivative (PD) controller have been applied to the same Baxter robot in this article to further show the superiority of our control scheme. Similar to the comparison simulation above, the parameters in (51)–(53) are set as  $k_1 = [10, 4]^T$ ,  $k_2 = [2, 2]^T$ ,  $\alpha_i = 1.5$ , and  $\gamma_i = 1$  for  $i = 1, 2$  according to a number of experiments we have made. And the PD controller is designed as  $\tau = -K_p e - K_d \dot{e}$ , where  $K_p = \text{diag}\{50, 40\}$  and  $K_d = \text{diag}\{\sqrt{0.5}, 0.5\}$ . The desired trajectory and the output constraints are all set the same as that in the above experiment study, and the initial position errors are all set the same as  $[0.1, -0.1]^T$  with the three controllers.

### B. Experimental Results

The experimental results are shown in Figs. 9–14. With four different initial conditions, we can see that the tracking error  $e_i$  and the FLS weight estimation  $\hat{W}_i$  can all achieve fixed-time convergence from Figs. 9, 10, and 12. Moreover, from Fig. 9, the

Fig. 11. Control input of left  $e_1$  and  $w_1$  joints with the proposed controller.Fig. 12. FLS weight estimation of  $W_1$  and  $W_2$  with the proposed controller.Fig. 13. Tracking error of left  $e_1$  and  $w_1$  joints with different controllers.

tracking errors of the left  $e_1$  and  $w_1$  joints can be both convergent to a small neighborhood around zero of  $[-0.05, 0.05]$  with the proposed controller, and the predefined transient performance  $-\beta_i < e_i < \beta_i$  can be guaranteed during the operation. The bounded control input  $\tau_i$  with different initial conditions is shown in Fig. 11. It can be seen that the input upper limit of the Baxter joints is not violated during the operation except for a short period of time in the initial moment with the initial errors  $[-0.8, 0.8]$ , which may due to the initial errors being too close to the limits of the output constraints  $\pm\beta_i$ .

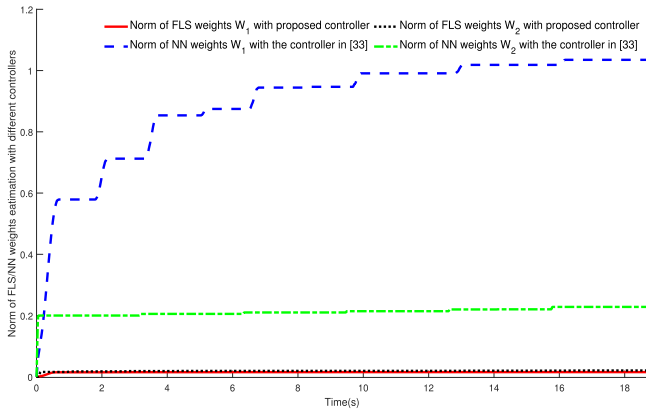


Fig. 14. FLS/NN weight estimation of the Baxter robot with the proposed controller and the controller in [33].

The comparison experimental results are shown in Figs. 13 and 14, from which we can see that all the controllers can achieve quite good tracking performance. In fact, better tracking performance can be achieved when more proper parameters are selected in [33] and the PD controller during our massive repeated experiment. However, when the selected parameters in the two comparative controllers are suboptimal, the predefined output constraints may be violated, such as the results in Fig. 13. However, the same predefined transient constraints can always be guaranteed with the proposed control scheme within the machine tolerance. Thus, the superiority of our work has been shown in terms of the transient performance. In addition, the weight estimation can always achieve faster convergence with the proposed control scheme in our massive repeated comparison experiment, such as the results in Fig. 14, which has shown the superiority of the proposed scheme from another level.

## VI. CONCLUSION

This article developed an adaptive fixed-time fuzzy control scheme for an uncertain robot system such that the system can achieve convergent in predefined time regardless of the initial conditions. A newly designed symmetric BLF was integrated into the controller design such that the desired transient performance can be guaranteed and the steady-state error can be artificially designed rather than constrained in the uncertain residual set in Lemma 1, like most practical fixed-time control schemes. A novel adaptive law was proposed such that the boundedness of all the closed-loop signals can be proved. Then, the assumption that the weight estimation is bounded in recent fixed-time control research [23]–[25] can be relaxed. The feasibility and superiority of the proposed controller were demonstrated by simulation and experimental results.

As is known, the common function approximation technique FLS/NN can only work over an unpredictable compact set, which has affected the system control performance. In the future, the fixed-time fuzzy control problem of unknown robots with global stability will be discussed.

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